1) (1) 3

2) (3) Infinitely many solutions

3) $\sqrt{(-6-0)^2+(8-0)^2}$

$=\sqrt{36+64}$

$\sqrt{100}$

$=10$

(1) 10 units

4) (2) proportional

(5) (1) $\frac{1}{2}$

6) (1) x^3y^3

64
36
100

$\sqrt{3}$ $\frac{\sin}{\cos} = 2\sin\theta$

$2\cos = \sqrt{3}$

$\cos\theta = \frac{\sqrt{3}}{2}$

$\sin^2 - \frac{3}{4} = \frac{1}{2}$

$\frac{2+3}{4} = \frac{5}{4}$

$\sqrt{\ }\ x^3y^3$

7) Let the roots be α & β, $\alpha+\beta = \frac{1}{2}\alpha\beta$

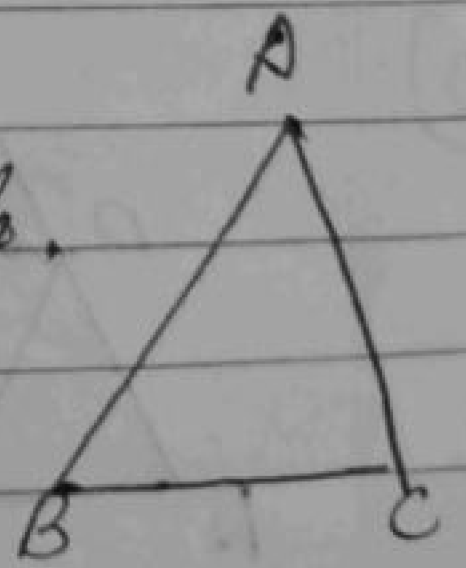
$\frac{-(-2)}{4} = \frac{1}{2} \times \frac{(K-4)}{4}$

$2 = \frac{1}{2} \times \frac{K-4}{4}$

$4 = \frac{K-4}{4}$

$16 = K-4$ $\boxed{K=20}$

8) ~~(0,0)~~ Let ABC be the equilateral triangle.



$A = (0,0)$

$B = (3,\sqrt{3})$

$C = (3,x)$

$AB = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$

$= \sqrt{(3-0)^2+(\sqrt{3}-0)^2}$

$= \sqrt{9+3}$

$= \sqrt{12}$

AB = BC So.

$BC\ \sqrt{(3-3)^2+(x-\sqrt{3})^2} = \sqrt{12}$

$(x-\sqrt{3})^2 = 12$

$x^2 - 2\sqrt{3}x + 3 = 12$

$x^2 - 2\sqrt{3}x - 9 = 0$

$\cancel{x^2 - 3\sqrt{3}x + \sqrt{3}x - 9}$

$\cancel{x(x-3\sqrt{3})\ 3x}$

$x^2 + \sqrt{3}x - 3\sqrt{3}x - 9$

$x(x+\sqrt{3}) - 3\sqrt{3}(x+\sqrt{3})$

$(x+\sqrt{3})\ (x-3\sqrt{3})$

$BC = (3\sqrt{3}, -\sqrt{3})$

$-9 < \sqrt{3},\ -3\sqrt{3}$

$-2\sqrt{3}$

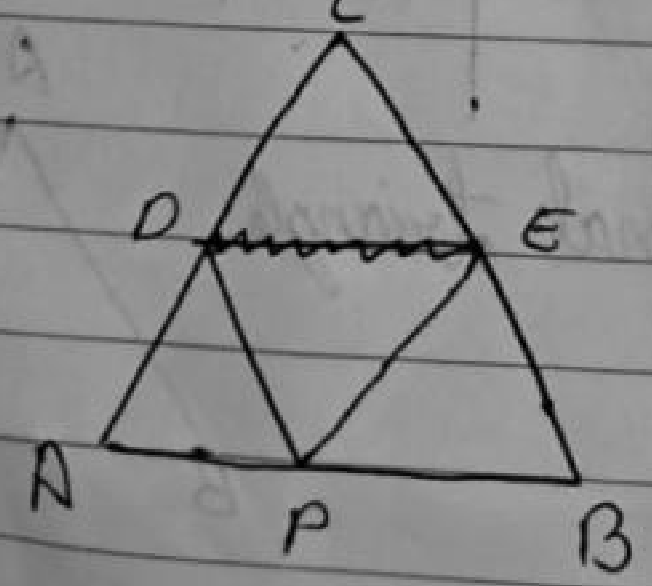
9) ~~2p~~ no. solution $= \frac{a_1}{a_2} \neq \frac{b_1}{b_2}$

So, $\frac{2p}{2} = \frac{3}{6}$

$= \frac{2p}{x} = \frac{1}{2}$

$p = \frac{1}{2}$

$\therefore$ p can be any value other than $\frac{1}{2}$ so it has only one solution.

10) 

PD ∥ BC

Given $\Rightarrow \frac{AD}{CD} = \frac{AP}{BP}$ [BPT theorem] -①

Also given $\Rightarrow \frac{AD}{CD} = \frac{CE}{BE}$ -②

From ① & ②

$\frac{AD}{CD} = \frac{AP}{BP} = \frac{CE}{BE}$

So $\frac{BE}{CE} = \frac{BP}{AP}$

$\therefore$ PE ∥ AC [Converse of BPT]

11)

15) $3x - (a+1)y = 2b-1$
$5x + (1-2a)y = 3b$

Infinitely many solutions $\rightarrow \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

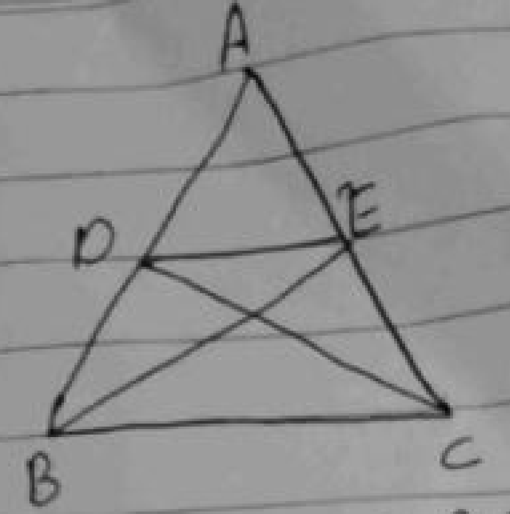
$\frac{3}{5} = \frac{-(a+1)}{1-2a} = \frac{2b-1}{3b}$

$\frac{3-6a-5a+5}{5-10a} = \frac{2b-1}{3b}$

$\frac{9-11a}{5-10a} = \frac{3}{5}$

$9-11a = 3$
$-11a =$

$3-6a = -5a+5$
$a+2=0$
$a=2$
$\frac{3-6a}{-5a+5} =$

16) 

Given $\triangle ABE \cong \triangle ACD$

i.e. $AB = AC$ [C.P.C.T]

~~In △~~ Also $AD = AE$ — ①

In $\triangle ADE$ & $\triangle ABC$

$\angle A = \angle A$ [common]

$AD = AE$ [From ①]

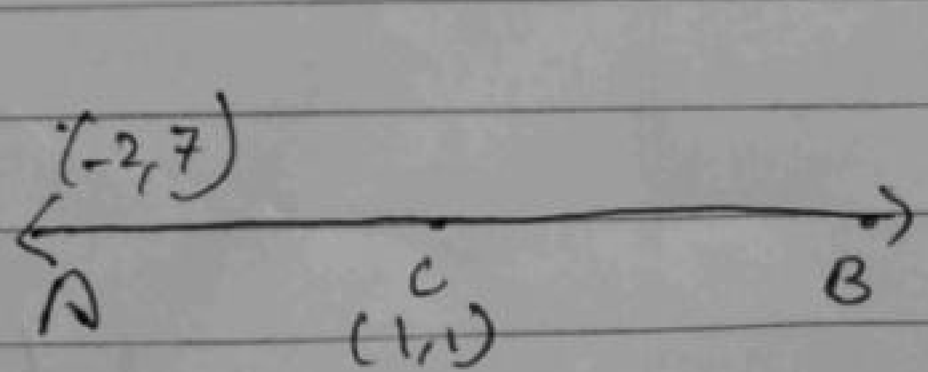
$\therefore \triangle ABC \sim \triangle ADE$ [SAS similarity]

17) $$\frac{\left(\frac{\sin\theta}{\cos\theta}\right)^2}{\sec^2\theta} + \frac{1 + \cot^2\theta}{1} =$$

$1 + \tan^2\theta$

$$\frac{\left(\frac{\sin\theta}{\cos\theta}\right)^2}{\sec^2\theta} + 1 - \frac{\cos\theta}{\sin\theta} = \frac{1}{\sin^2\theta - \cos^2\theta}$$

18) Section ~~Midpoint~~ formula $\rightarrow \left(\frac{m_1x_2 + m_2x_1}{m_1+m_2}, \frac{m_1y_2 + m_2y_1}{m_1+m_2}\right)$



$$\left(\frac{3x_2 + 2(-2)}{3+2}, \frac{3y_2 + 2(7)}{3+2}\right)$$

$$\frac{3x_2 - 4}{5} = 1$$

$3x_2 - 4 = 5$

$3x_2 = 9$

$x_2 = \frac{9}{3}$

$\boxed{x_2 = 3}$

$\frac{3y_2+14}{5}=1$

$3y_2+14=5$

$3y_2=5-14$

$3y_2=-9$

$\boxed{y_2=-3}$

Therefore $B=(3,-3)$

19) Let the Numerator & denominator be x & y respectively.

$\frac{2x}{y+1}=1$

$2x=y+1$

$2x-y-1=0$ — ①

Case ②

$\frac{x+4}{y(2)}=\cancel{1}$

$=x+4=2y$

$=x-2y+4=0$ — ②

①-②

$2x-y\cancel{+}=\cancel{0}$ -1

$-\cancel{2}x-4y\cancel{+4}=\cancel{0}$ 4

$3y=-5$

$\boxed{y=\frac{-5}{3}}$

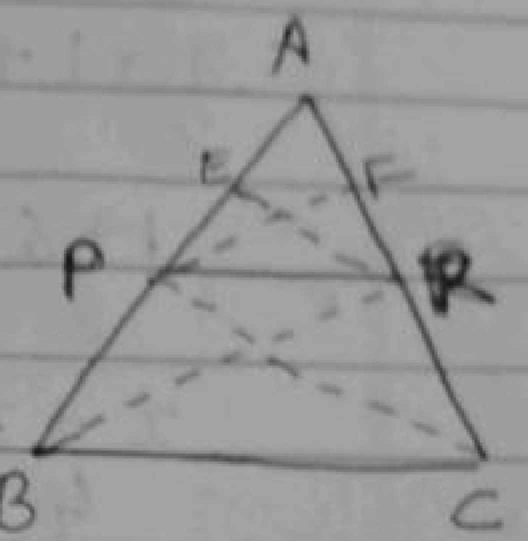
Substitute $y=\frac{-5}{3}$ in eq ②

$x-2\left(\frac{-5}{3}\right)=4$

$x-\frac{10}{3}=4$

$x=\frac{4}{}+\frac{10}{3}$ $\quad x=\frac{12+10}{3}=\boxed{\frac{22}{3}}$

20) Basic proportionality theorem →
If a line drawn parallel to one side of the triangle



Given

$PR \parallel BC$

To prove

$$\frac{AP}{BP} = \frac{AR}{CR}$$

Construction → Draw PF & ER
Draw BR & PC

Proof ~~In~~ $\triangle$ $Ar.\ \triangle\ \overset{APR}{\cancel{PAF}} = \frac{1}{2} \times \overset{AP}{\cancel{AR}} \times \overset{ER}{\cancel{PR}}$ — ①

$Ar.\ \triangle\ \overset{BPR}{\cancel{AER}} = \frac{1}{2} \times \overset{BP}{\cancel{AE}} \times \overset{ER}{\cancel{PR}}$ — ②

$$\frac{①}{②} \rightarrow \frac{\cancel{1/2} \times AP \times \cancel{ER}}{\cancel{1/2} \times BP \times \cancel{ER}}$$

$$= \frac{AP}{BP} \quad — ③$$

$$Ar.\ \triangle ARP = \frac{1}{2} \times AR \times PF \quad — ④$$

$$Ar.\ \triangle PRC = \frac{1}{2} \times CR \times PF \quad — ⑤$$

$$\frac{④}{⑤} \rightarrow \frac{\cancel{1/2} \times AR \times \cancel{PF}}{\cancel{1/2} \times CR \times \cancel{PF}} = \frac{AR}{CR} \quad — ⑥$$

From ③ & ⑥ $\frac{AP}{BP} = \frac{AR}{CR}$

21) $\dfrac{\sin 30^\circ + 2\cos^2 45^\circ + \tan^2 60^\circ}{\frac{1}{2}\cot 45^\circ + \cos^2 30^\circ + \tan^2 45^\circ}$

$0, \frac{1}{2}, \frac{1}{\sqrt{2}}, \frac{\sqrt{3}}{2}, 1$

$1, \frac{\sqrt{3}}{2}, \frac{1}{\sqrt{2}}$

$0, \frac{1}{\sqrt{3}}, 1, \sqrt{3},$

$$\dfrac{\frac{1}{2} + 2 \times \frac{1}{2} + 3}{\frac{1}{2} \times 1 + \frac{3}{4} + 1^2}$$

$$= \dfrac{\frac{1+6}{2}}{\frac{2+3}{4} + 1}$$

$$= \dfrac{\frac{7}{2}}{\frac{5}{4} + 1}$$

$$= \dfrac{\frac{7}{2}}{\frac{5+4}{4}}$$

$$= \dfrac{\frac{7}{2}}{\frac{9}{4}}$$

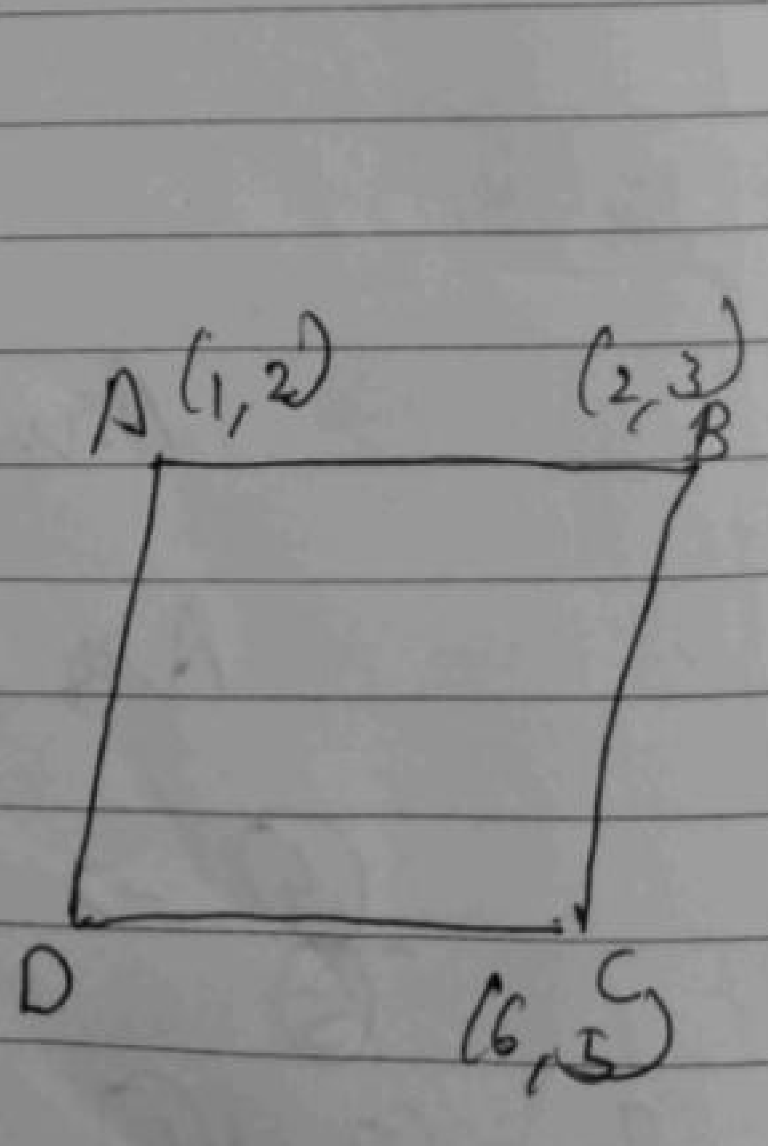
$$= \frac{7}{\cancel{2}} \times \frac{\cancel{4}^{\,2}}{9}$$

$$= \boxed{\frac{14}{9}}$$

22) AB = CD [opposite sides]

Distance formula

$= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$



$AB = \sqrt{(2-1)^2 + (3-2)^2}$

$= \sqrt{1+1}$

$= \sqrt{2}$

$CD = \sqrt{(x_2-6)^2+(y_2-5)^2} = \sqrt{2}$

$(x_0-6)^2+(y_0-5)^2 = 2$

$x^2 + 12x + 36 + 25 + y^2 + 10y \cancel{+25} = 2$

$x^2+y^2+61+12x+10y = 2$

23) Lcm of 330, 198 & 220

$330 = 10 \times 3 \times 11 = 5 \times 2^1 \times 3 \times 11$

10	330
3	33
11	11
	1

$198 = 2 \times 3^2 \times 11$

2	198
3	99
3	33
11	11
	1

$220 = 2^2 \times 5 \times 11$

10	220
2	22
11	11
	1

$Lcm(330, 198, 220) = 5^1 \times 2^2 \times 11 \times 3^2$

$= 5 \times 4 \times 11 \times 9$

$= 1980$ min.

After 33 hours they will be together.

180×11

180
180
1980

$1980 \div 60 = 33$

24) Let the father's age be x & son's age by y.

$2y + x = 70 \quad -①$
$2x + y = 95 \quad -②$

$x + 2y = 70$
$2x + y = 9\cancel{6}5$

Adding eq ① by ~~②~~ 2

$2x + 4y = 140$
$-\ 2x + y = \cancel{70}\ 95$

$3y = \cancel{70}\ 45$

$y = \cancel{\frac{70}{3}}\ 23 \ \cancel{y} \quad 3 \cdot y = \frac{\cancel{45}}{3}\ 15$

$\boxed{y = \cancel{7}15}$

Substituting $\boxed{y = \cancel{7}5}$ in eq ②

~~$2(35) + x = 70$~~
~~$70 +$~~ $\quad 2x + \cancel{35}\ 15 = 95$

$2x = \cancel{60}$

$\boxed{x = \cancel{7}0}$

95
$-\ \cancel{3}5$
$\cancel{8}0$

Father's age → ~~35~~ 40
Son's age → ~~20~~ 15

25) Section formula

A $(1, -7)$ $\quad$ ~~1 : y~~ $x : 1$ $\quad$ B $(6, 4)$

$\rightarrow \left(\frac{m_2 x_1 + m_1 x_2}{m_1 + m_2}, \frac{m_2 y_1 + m_1 y_2}{m_1 + m_2} \right)$

$\rightarrow \left(1(1) + x(\right.$

26) cosec c θ
((c

27) (i) 3

(ii) 15 mil

(iii) ~~16 m~~

(iv) ~~10 m~~

28) (i)

26) $cosec\theta - \sin\theta = m,\ \sec\theta - \cos\theta = n$

$\left((cosec\theta - \sin\theta)^2 (\sec\theta - \cos\theta) \right)^{2/3} +$

27) (i) 3:2

(ii) 15 miles

(iii) ~~16 miles~~ $\frac{10}{7}$ ~~km~~ miles.

(iv) ~~10 miles~~ 3.5 miles

28) (i) Let (x_1, y_1) be $(2,5)$
(x_2, y_2) be $(8,3)$
ratio → 1:2
Section formula →

$$\left(\frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}, \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2} \right)$$

$$= \left(\frac{1(8) + 2(2)}{1+2}, \frac{1(3) + 2(5)}{3} \right)$$

$= \left(\frac{\cancel{12}^{4}}{\cancel{3}}, \frac{13}{3}\right)$

$= \left(4, \frac{13}{3}\right)$

(ii) Distance formula $\rightarrow \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$

Let (x_1, y_1) be $(2,5)$

(x_2, y_2) be $(8,3)$

$\sqrt{(8-2)^2+(3-5)^2}$

$\sqrt{36+4}$

$\sqrt{40}$ units

(iii) 2units

(iv) Midpoint formula $\rightarrow \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$

$= \left(\frac{2+8}{2}, \frac{5+3}{2}\right)$

$= \left(\frac{\cancel{10}^{5}}{\cancel{2}}, \frac{\cancel{8}^{4}}{\cancel{2}}\right)$

$= (5,4)$

11) $\tan\theta = \frac{1}{2}$

$= \frac{\sin\theta}{\cos\theta} = \frac{1}{2}$

$= \sin\theta = \frac{1}{2}.\cos\theta$

$= \sin^2\theta = \frac{1}{2}.\cos\theta.\sin\theta$

$\tan\theta = \frac{Opp}{Adj.}$

$1^2 + 2^2 = H^2$

$1+4 = H^2$

$5 = H^2$

$H = \sqrt{5}$

$\left(\frac{1}{\sqrt{5}}\right)^2 =$

$\frac{1}{2} \times$